CHEAT SHEET · JS INTERVIEW #03 @specsycoder

Shallow vs Deep Copy

Why the original changed

```
const user = { name: "Aman", address: { city: "Delhi" } };
```

snippet-1.js

```
1 const copy = { ...user };
2 copy.address.city = "Pune";
3 console.log(user.address.city);
```

output **Pune**

snippet-2.js

```
1 const copy = structuredClone(user);
2 copy.address.city = "Pune";
3 console.log(user.address.city);
```

output **Delhi**

The 4 ideas

01 Reference
An object variable stores WHERE the object lives in memory, not the object itself.

02 Shallow copy
New outer object, but nested objects are shared by reference.

03 Deep copy
Copies every level. Nothing is shared with the original.

04 Why it matters
Shared nested data causes state bugs in React, Angular and caches.

Step by step (snippet 1)

1. `{ ...user }` makes a new object with `name` and the SAME `address` reference.
2. `copy.address.city = "Pune"` edits that shared address object.
3. `user.address` points to the same object, so it now reads **"Pune"**.

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Which copy do I use?

Shallow tools vs deep tools

copies.js

```
1 const a = [...arr]; // shallow
2 const b = Object.assign({}, obj); // shallow
3 const c = structuredClone(obj); // deep
4 const d = JSON.parse(JSON.stringify(obj)); // deep*
```

method	depth	watch out
`{ ...obj } / [...arr]`	shallow	nested objects shared
`Object.assign()`	shallow	same as spread
`structuredClone()`	**deep**	throws on functions and DOM nodes
`JSON.parse(JSON.stringify())`	**deep***	Dates become strings; undefined and functions dropped

Traps to mention

- **Spread on an array of objects is still shallow.** The objects inside are shared.
- **Update nested state without deep cloning:** `{ ...user, address: { ...user.address, city: "Pune" } }`
- **const does not freeze objects.** It only stops reassigning the variable.

Your 20-second answer

"Objects are stored by reference. Spread and Object.assign make a shallow copy: a new outer object, but nested objects are still shared, so changing them changes the original. structuredClone makes a deep copy of every level, so the original stays safe."